

Knowledge Engineering (SCOM7348)

Subtype Relations & Other Constraints

(Chapter 6)

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Conceptual Schema Design Steps

1. From examples to elementary facts



2. Draw fact types and apply population check



3. Combine entity types



4. Add uniqueness constraints



5. Add mandatory constraints



6. Add subtype relations and other constraints



7. Final checks, & schema engineering issues



Outline

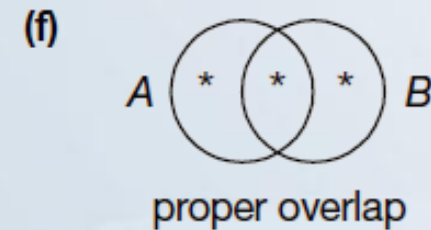
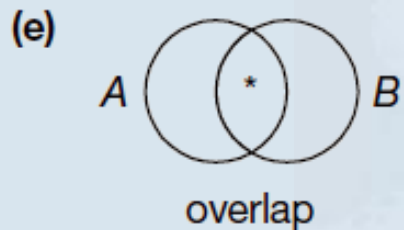
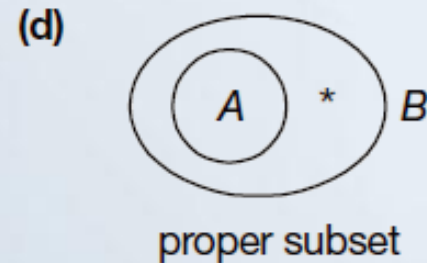
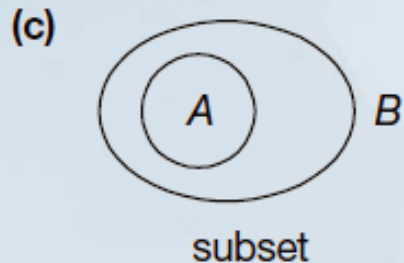
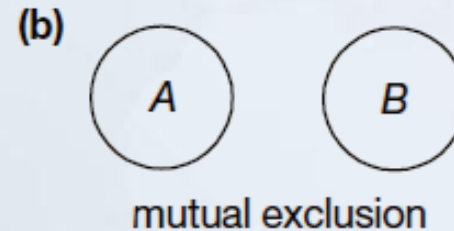
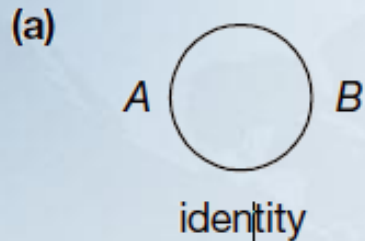
Quick Math background

- Value Constraints
- Set Constrains
 - Subset
 - Equality
 - Exclusion
- Subtype relations
- Frequency constraints

Information adapted from [1]

Mathematical Background

Hypothetical Euler diagrams for set comparisons.

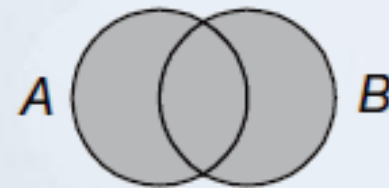


Mathematical Background

Venn diagrams for three set-forming operations.

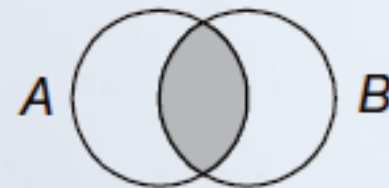
union

$$A \cup B$$



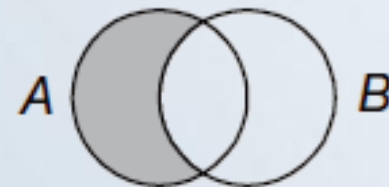
intersection

$$A \cap B$$

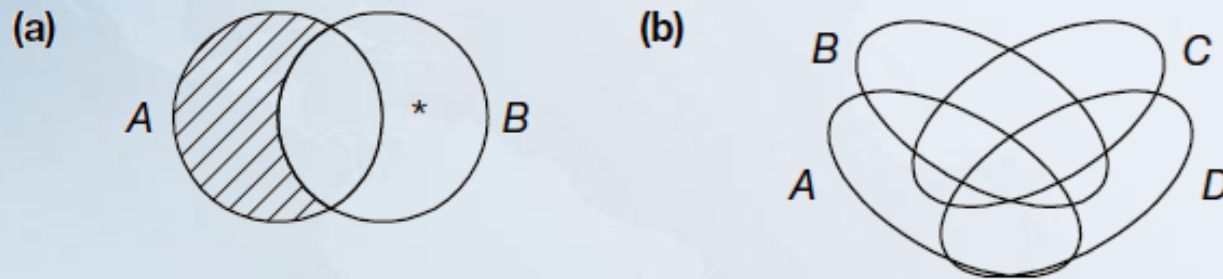


difference

$$A - B$$



Mathematical Background



Venn diagrams for (a) A is a proper subset of B and (b) four sets.

Outline

- Quick Math background

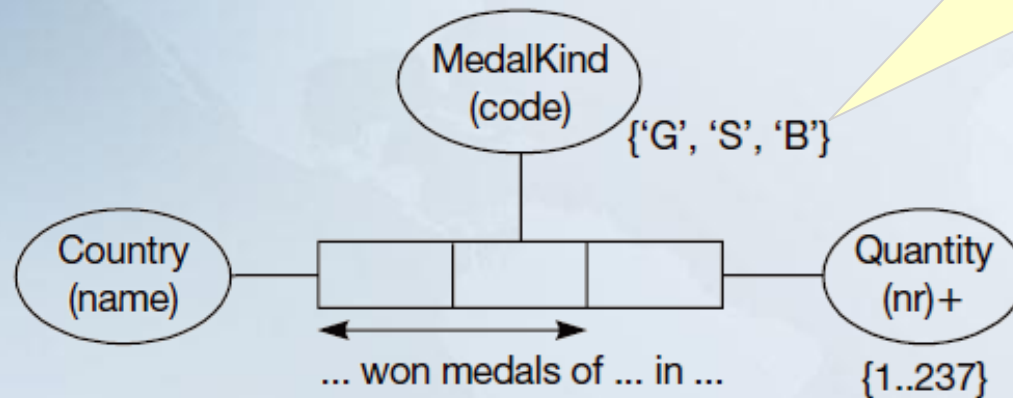


Value Constraints

- Set Constrains
 - Subset
 - Equality
 - Exclusion
- Subtype relations
- Frequency constraints

Value Constraint

Called **Value Constraint**

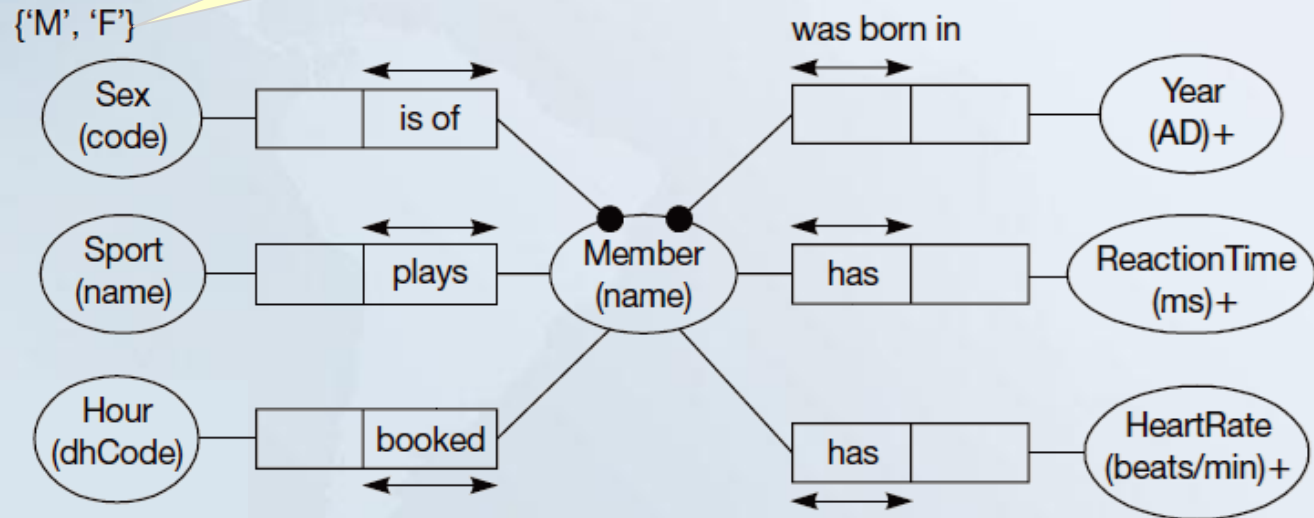


A set of values, from which the value of the MedalKind is limited to

Argentina	S	1
Argentina	B	1
Australia	G	3
Australia	S	6
Australia	B	5
...	...	...

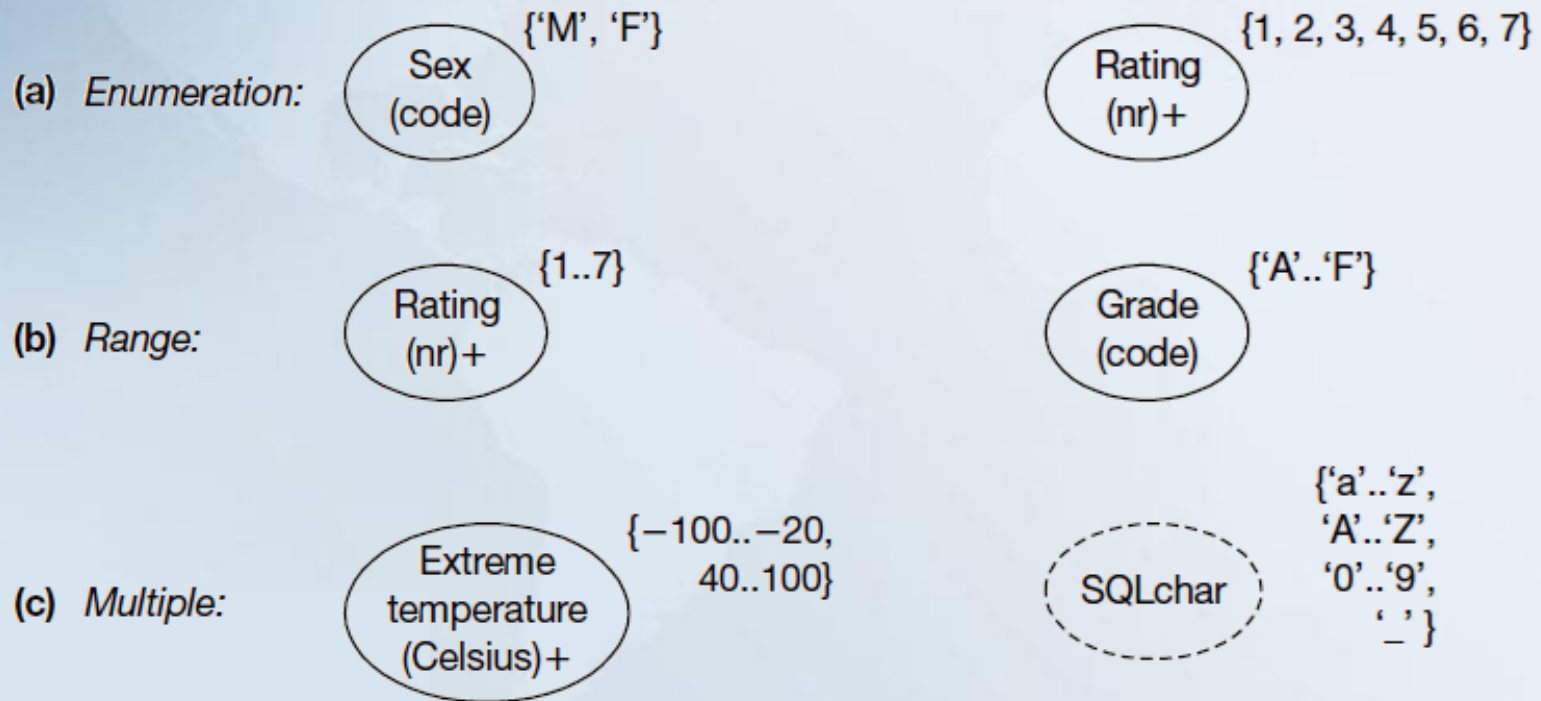
Value Constraint

The value of sex should be one of {'M', 'F'}



Value Constraint

Value constraints may list the possible values of a value type.

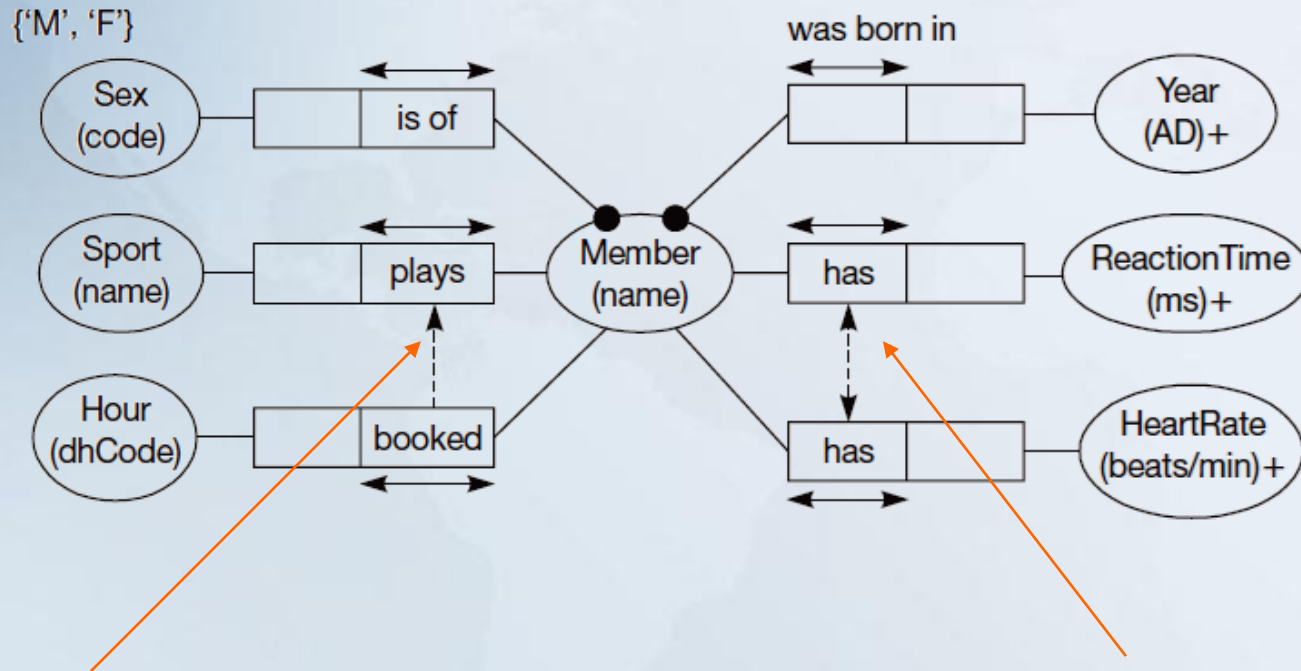


→ Who can give more examples?

Outline

- Quick Math background
- Value Constraints
-  Set Constrains
 - Subset
 - Equality
 - Exclusion
- Subtype relations
- Frequency constraints

Role subset/equality constraint



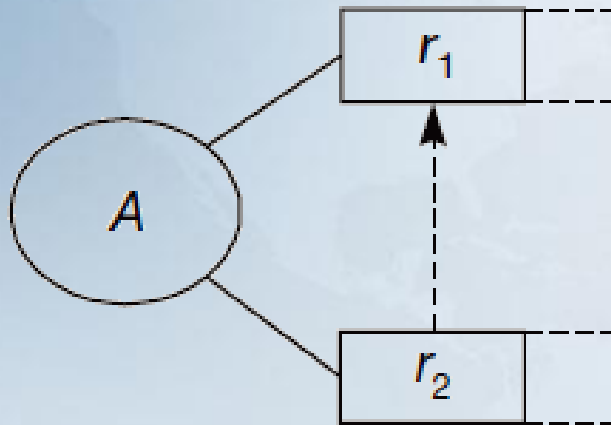
Subset constraint:

Every Member booked an Hour
should play sport.

Equality constraint:

Every Member 'has' ReactionTime
should 'has' HeartRate, and every
Member 'has' HeartRate should
'has' ReactionTime.

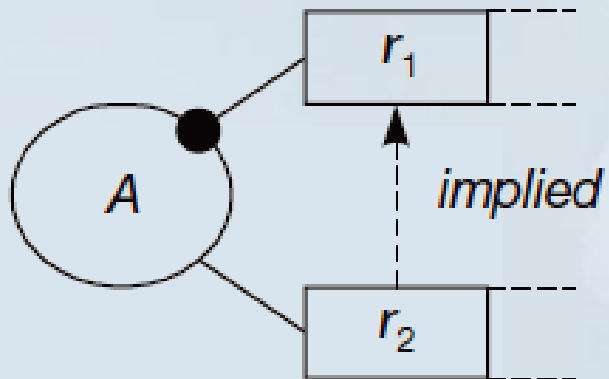
Role subset constraint



For each database state:

$$\text{pop}(r_2) \subseteq \text{pop}(r_1)$$

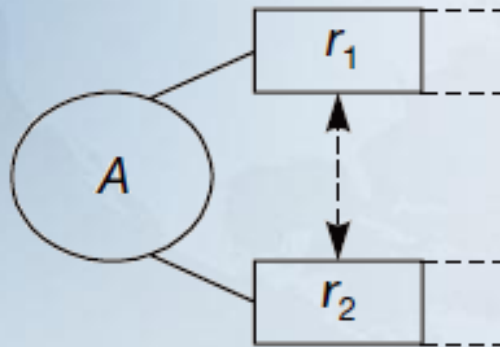
if a plays r_2 then a plays r_1



Notice that this subset constraint is implied, and should be removed.

That is, there is no need to say that every A playing r_2 must also play r_1 (subset), because the mandatory constraint here means that every A must play r_1 (the Mandatory implies the subset).

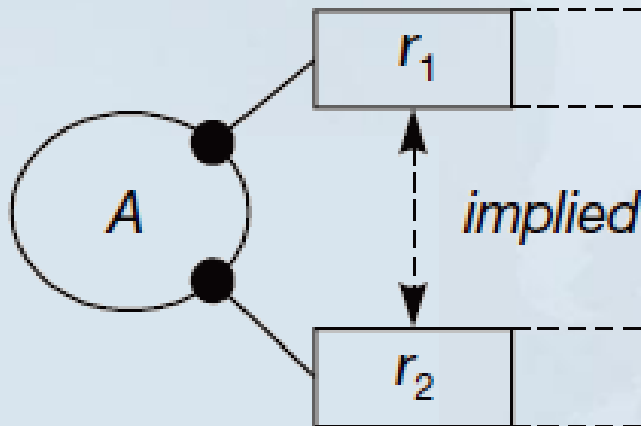
Role equality constraint



For each database state:

$$\text{pop}(r_1) = \text{pop}(r_2)$$

a plays r_1 if a plays r_2



**Also this quality constraint is implied,
and should be removed.**

Implication

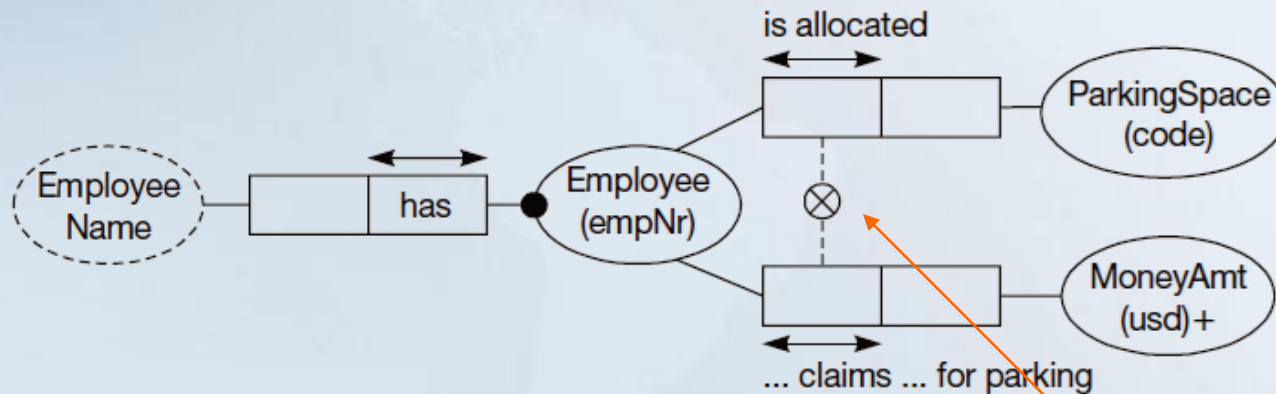
Who can explain the difference?



The two constraints in the first model says: each A must play r_1 or r_2 (or both), and that if A plays r_2 then it must play r_1 . This means that r_1 must be always played (which is the second model)

Role Exclusion Constraint

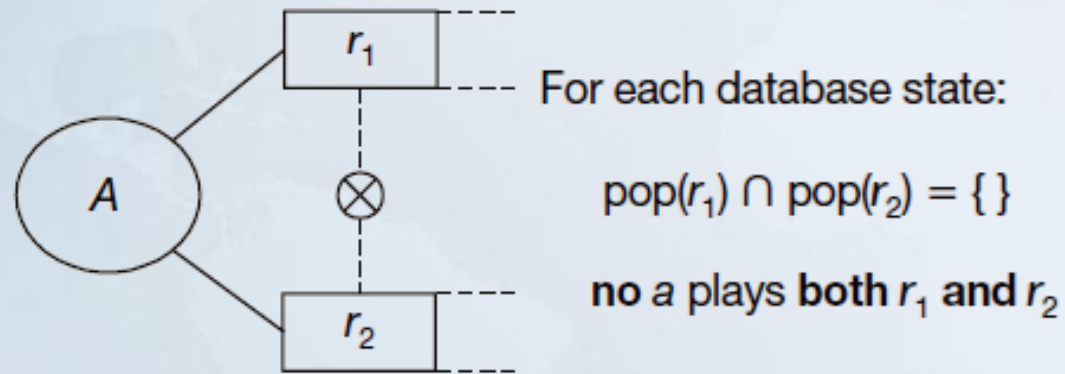
<i>EmpNr</i>	<i>Employee name</i>	<i>Parking space</i>	<i>Parking claim (\$)</i>
001	Adams B	C01	—
002	Bloggs F	—	200
003	Collins T	B05	—
004	Dancer F	—	250
005	Eisai Z	?	?



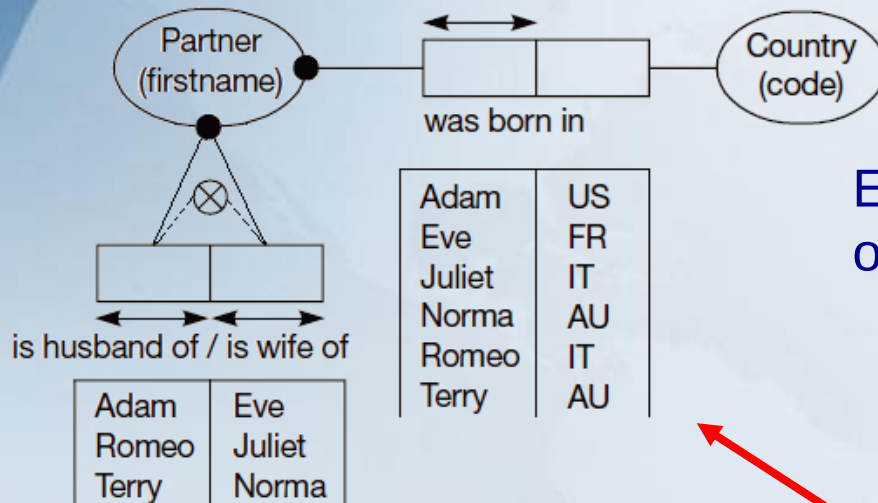
Exclusion constraint:

Every Employee is allocated a ParkingSpace **should not** claim MoneyAmt.

Role Exclusion Constraint

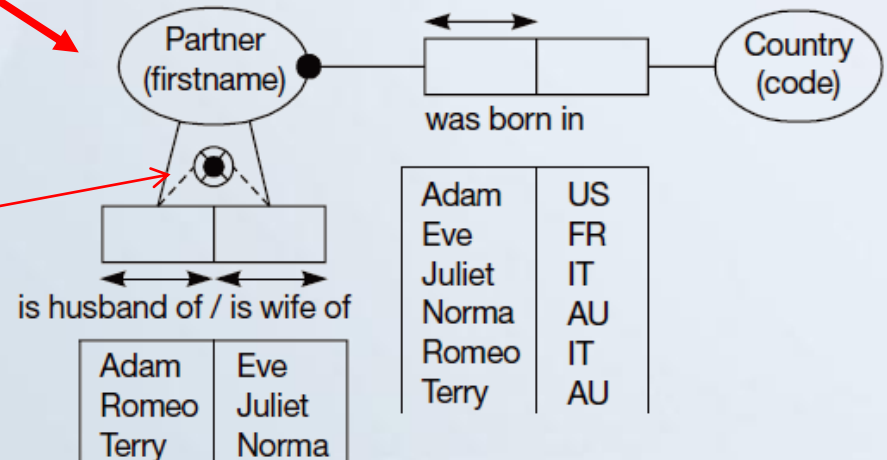


Role Exclusion Constraint

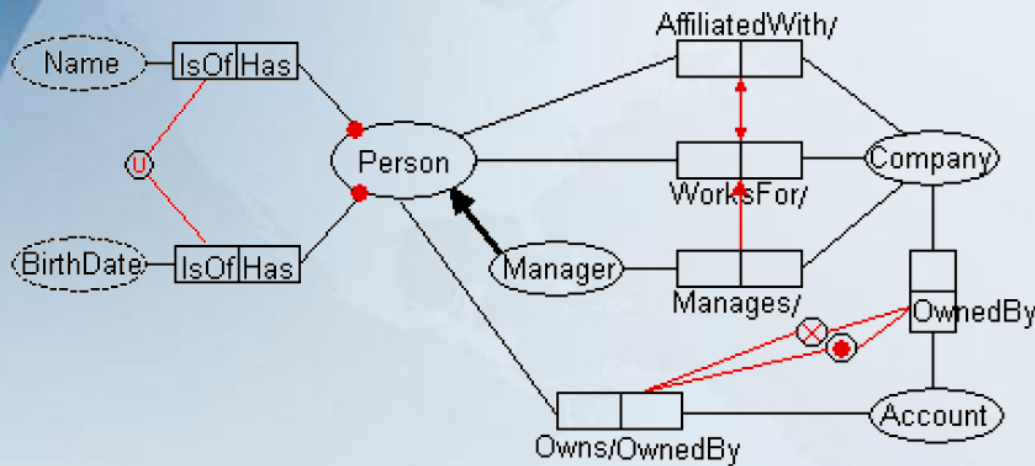


Each partner *must* be either a husband or wife (but not both at the same time).

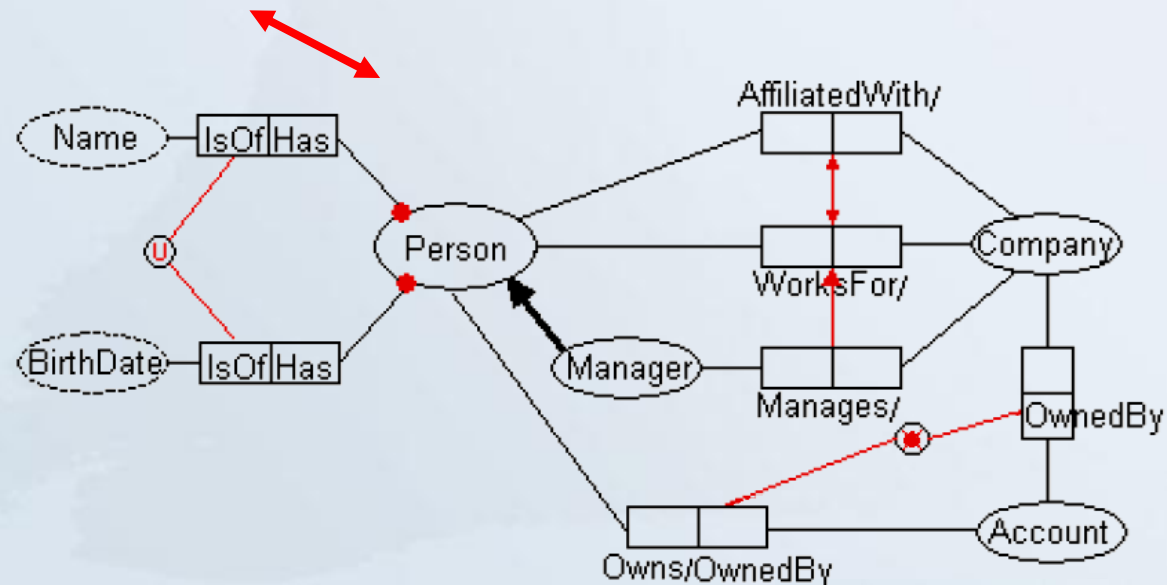
Called "*Exclusive-or*"



Exclusive-or (another example)

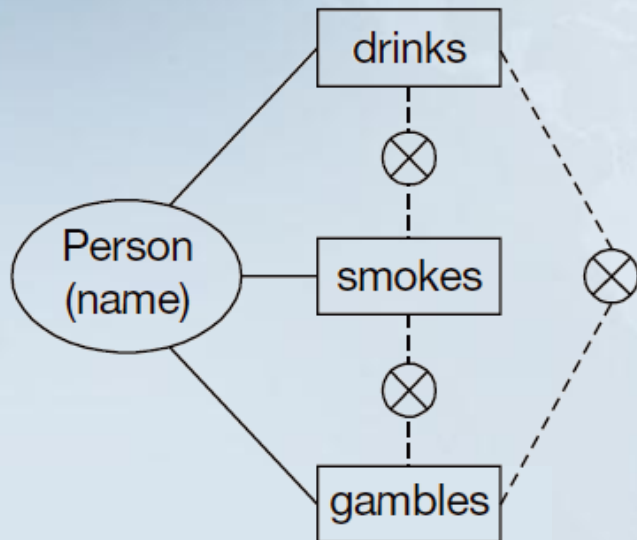


Each Account must be OwnedBy a Person or a Company, but not both.

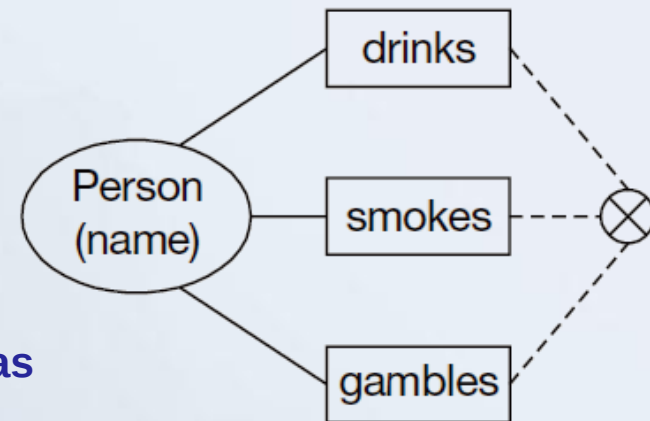


Role Exclusion Constraint

Each person has at most one of three vices. i.e., from 0 to 3 vices.



It can be written also as

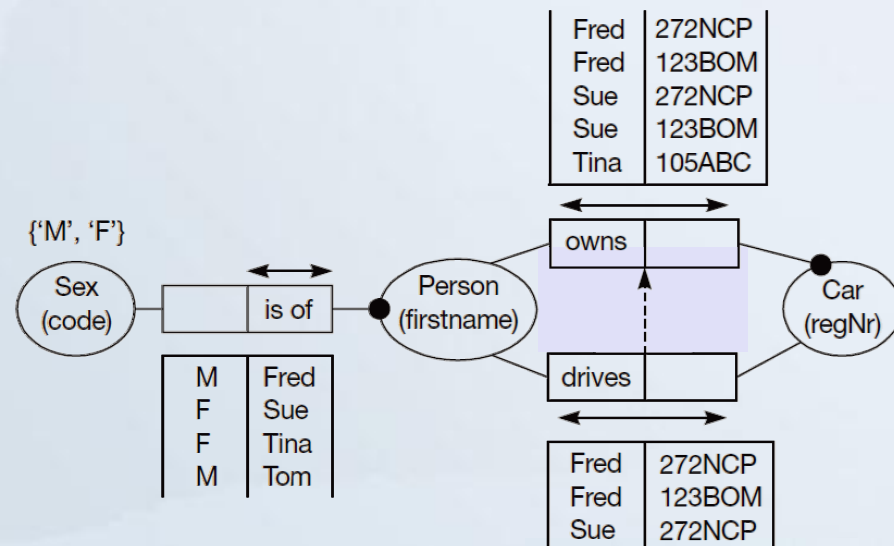


Pair Exclusion Constraint

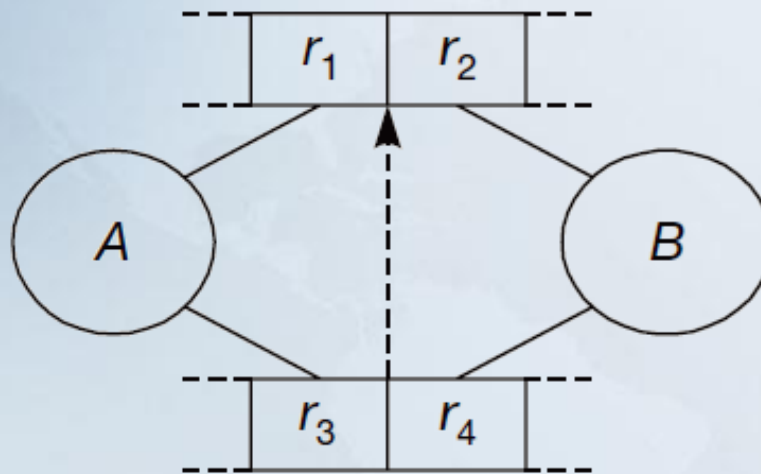


Person	Sex	Cars owned	Cars driven
Fred	M	272NCP, 123BOM	272NCP, 123BOM
Sue	F	272NCP, 123BOM	272NCP
Tina	F	105ABC	
Tom	M		

How can we restrict that a person can drive a car only if he owns *that* car.



Pair-subset constraint

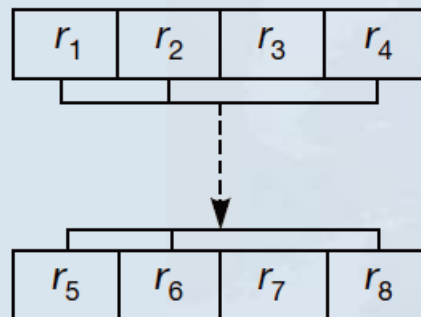


For each database state:

$$\text{pop}(r_3, r_4) \subseteq \text{pop}(r_1, r_2)$$

each *ab* pair in $\text{pop}(r_3, r_4)$ is also in $\text{pop}(r_1, r_2)$

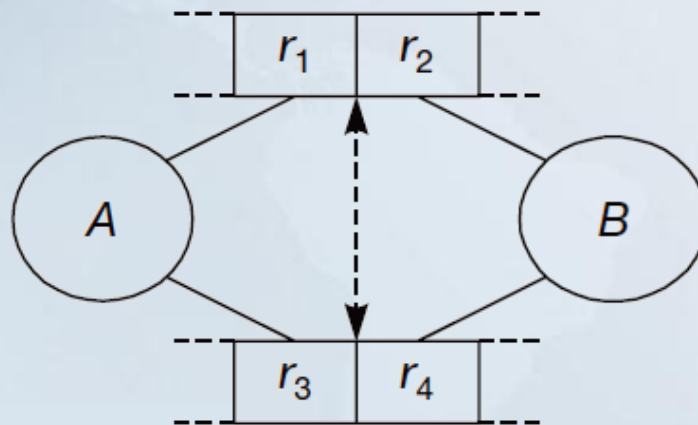
An example of a tuple-subset constraint between sequences of three roles.



For each database state:

$$\text{pop}(r_1, r_2, r_4) \subseteq \text{pop}(r_5, r_6, r_8)$$

Equality Constraint

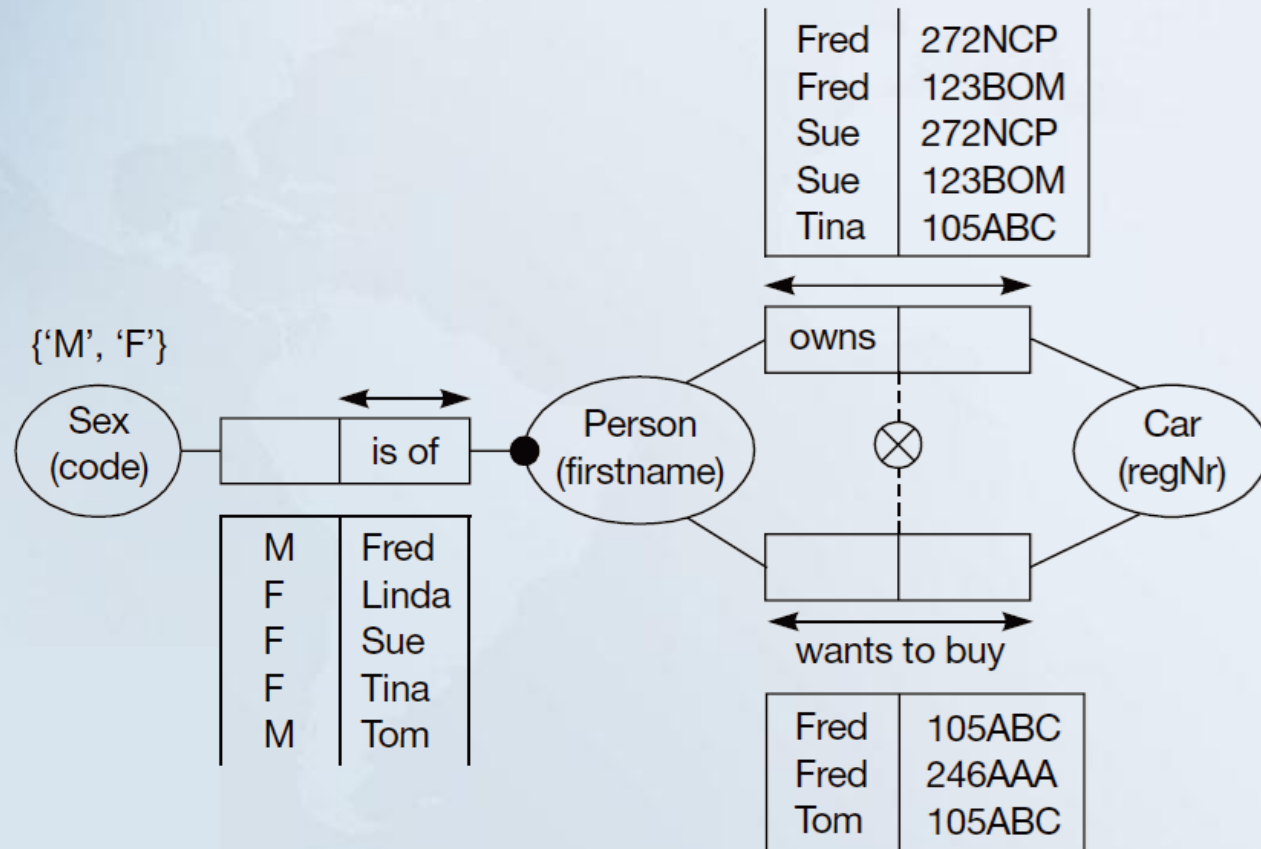


For each database state:

$$\text{pop}(r_3, r_4) = \text{pop}(r_1, r_2)$$

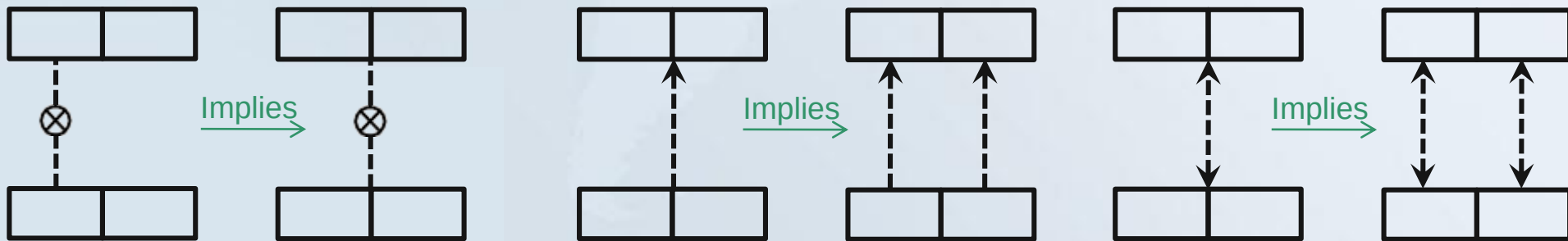
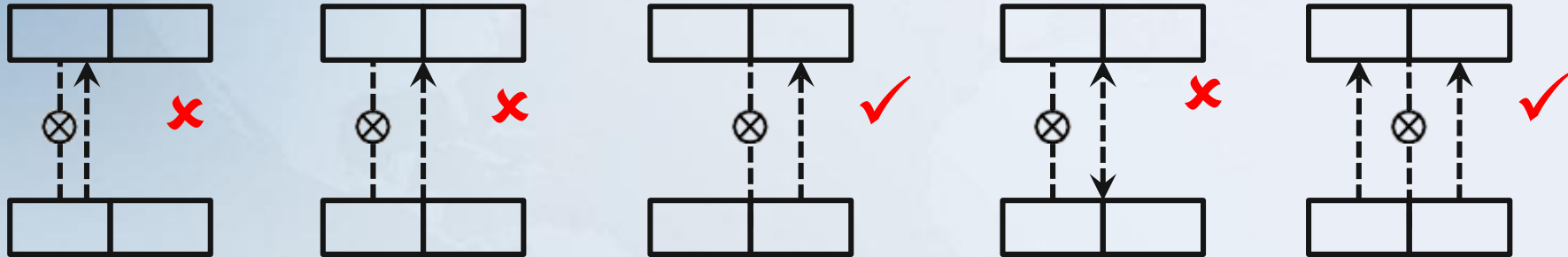
**each ab pair in $\text{pop}(r_3, r_4)$
is also in $\text{pop}(r_1, r_2)$
and vice versa**

Pair Exclusion Constraint



Same person can 'own' and 'wants to buy' the same car?

What is Wrong?



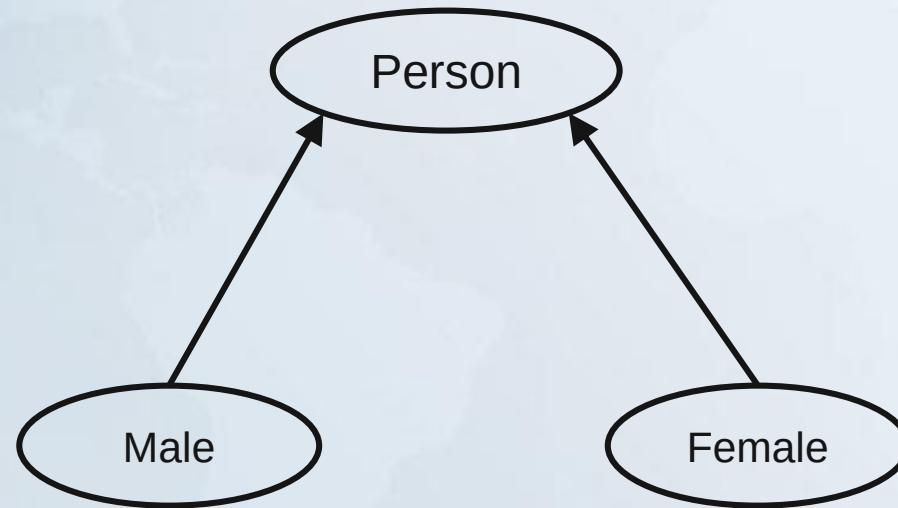
Outline

- Quick Math background
- Value Constraints
- Set Constrains
 - Subset
 - Equality
 - Exclusion

Subtype relations

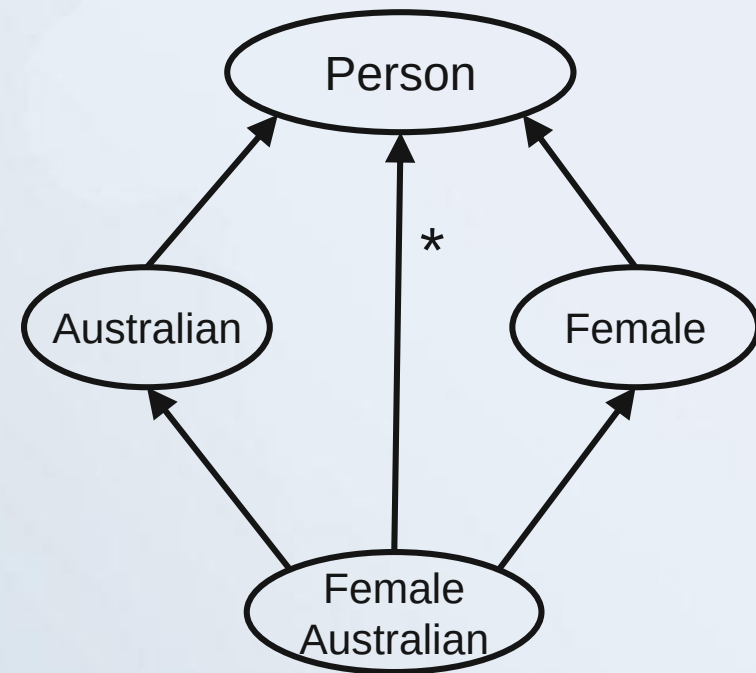
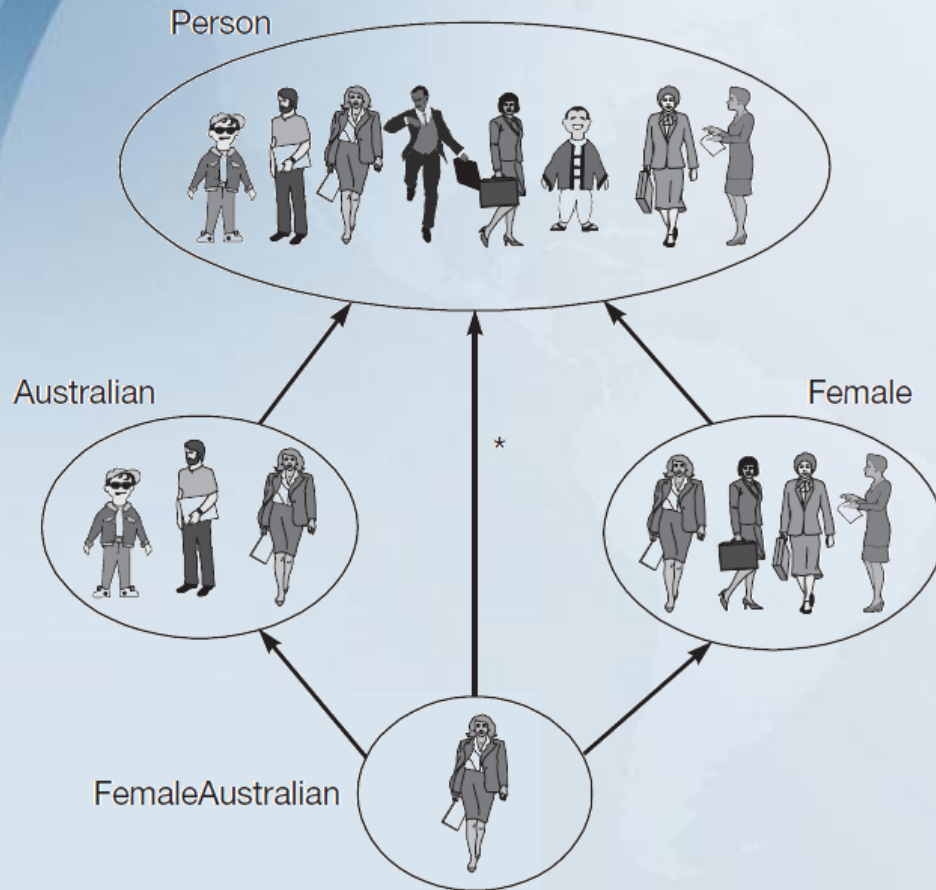
- Frequency constraints

Subtypes



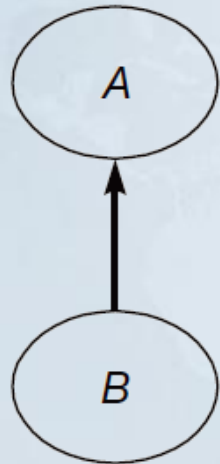
- Generalization/Specialization hierarchy.
- Subtype inherits the properties of its supertype.

Subtypes



* The indirect subtype connection is implied, so it should be omitted

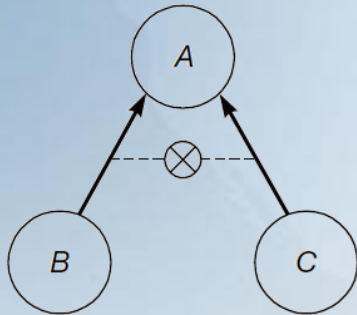
Subtypes



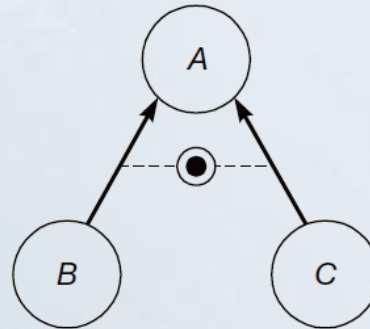
$A \neq B$, and for each database state:

$$\text{pop}(B) \subseteq \text{pop}(A)$$

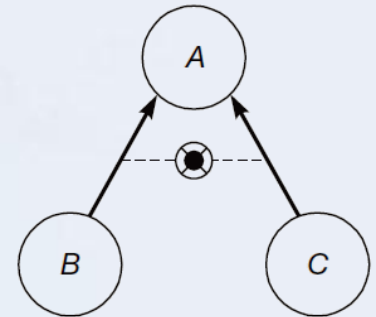
Subtypes



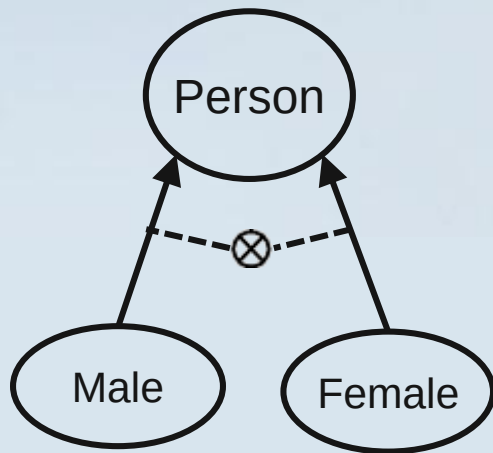
$$B \cap C = \{\}$$



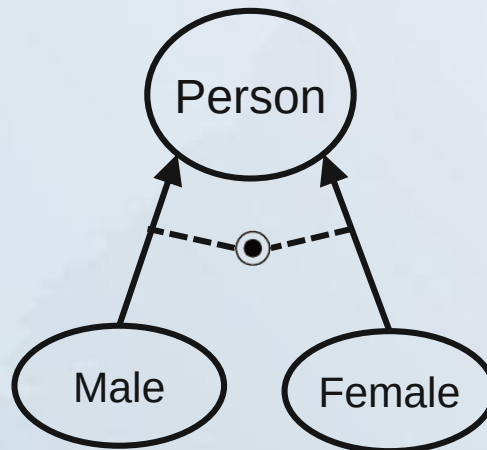
$$A = B \cup C$$



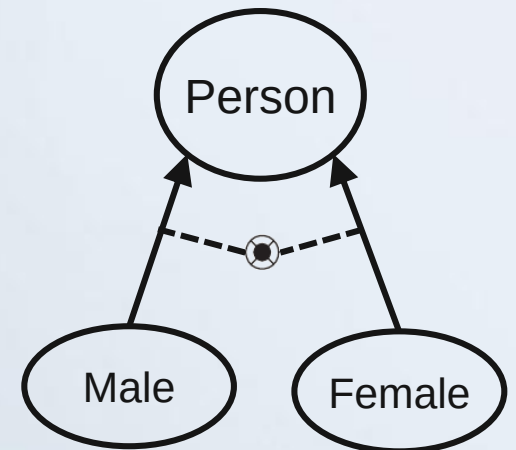
Both



There is no person that can be Male and Female at the same time.

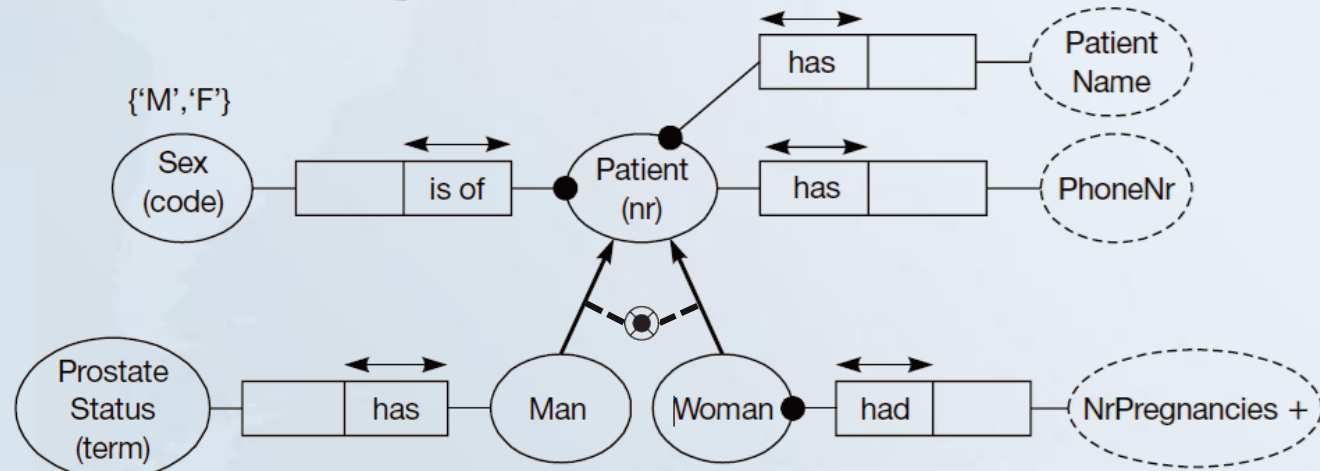
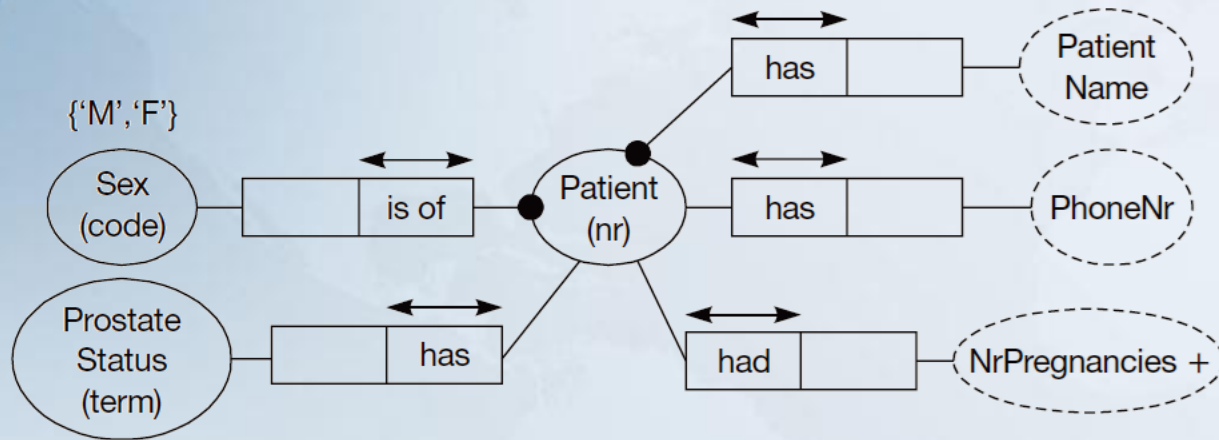


Every person must be a Male or a female.



Every person must be either a Male or a Female

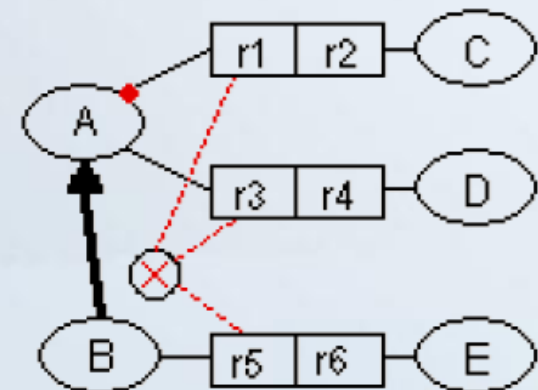
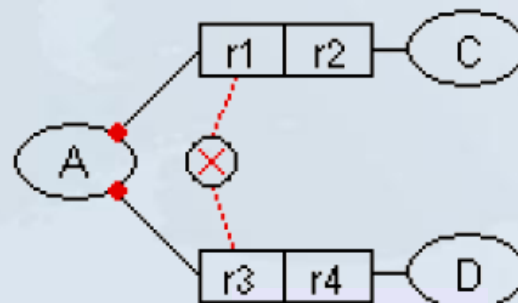
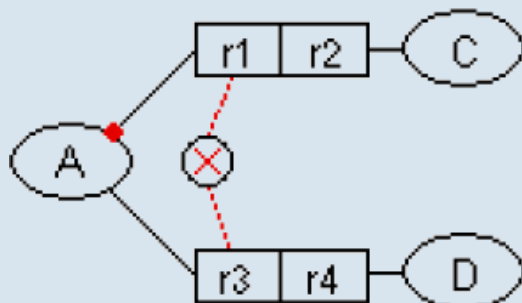
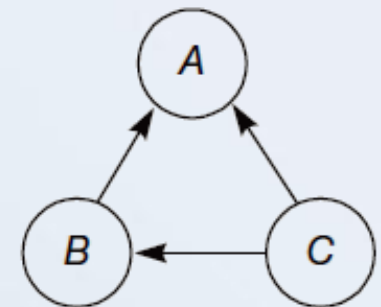
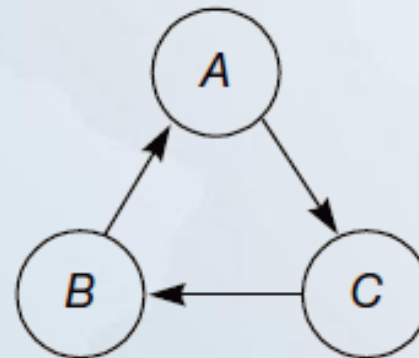
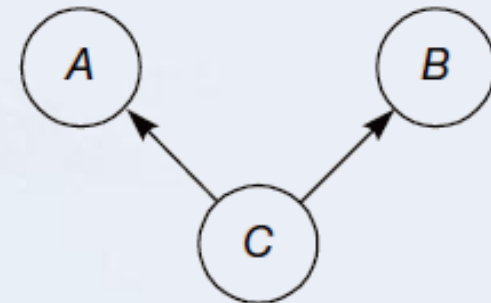
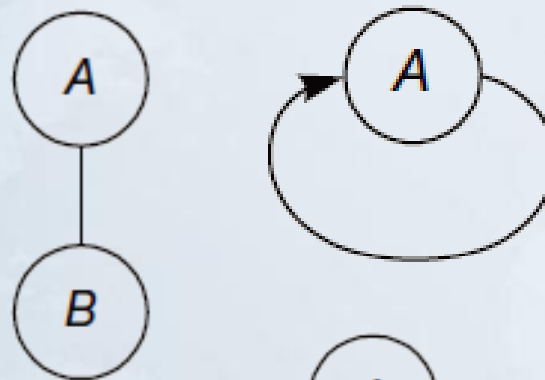
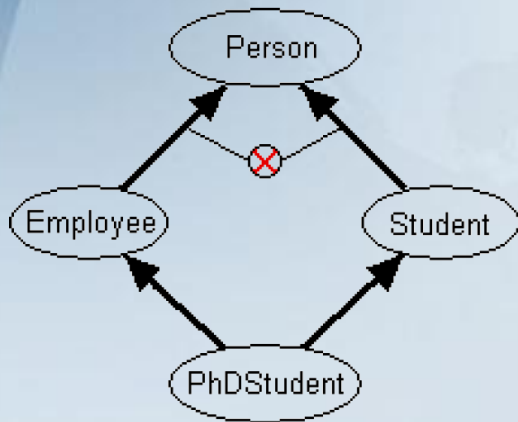
Subtypes



What is
Inherited?

each Man is a Patient who is of Sex 'M'
each Woman is a Patient who is of Sex 'F'

What is Wrong?



Outline

- Quick Math background
- Value Constraints
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 - Equality
 - Exclusion
- Subtype relations



Frequency constraints

also called “Occurrence constraints”

Frequency constraints

To indicate that each entry in a fact column must occur there exactly n times, the number n is written beside the role.

Each city in the first column must occur three times.

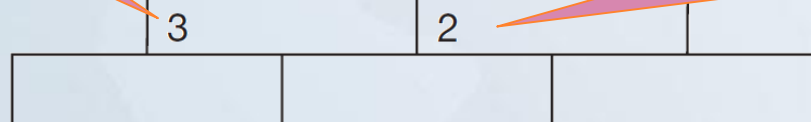
{'LA', 'Tokyo'} {'F', 'H', 'T'}

City
(name)

DriveKind
(code)

Quantity
(nr)+

each drive kind in the Second column must appear there twice

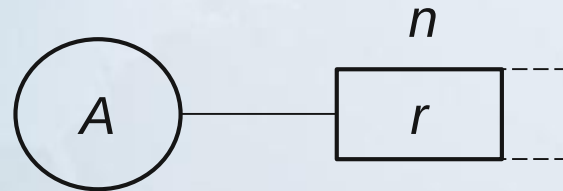


... has stocks of ... in ...

LA	F	3000
LA	H	2000
LA	T	600
Tokyo	F	2000
Tokyo	H	1800
Tokyo	T	300

A compound transaction is needed to initially populate this fact type requiring at least six facts to be added.

Frequency constraints



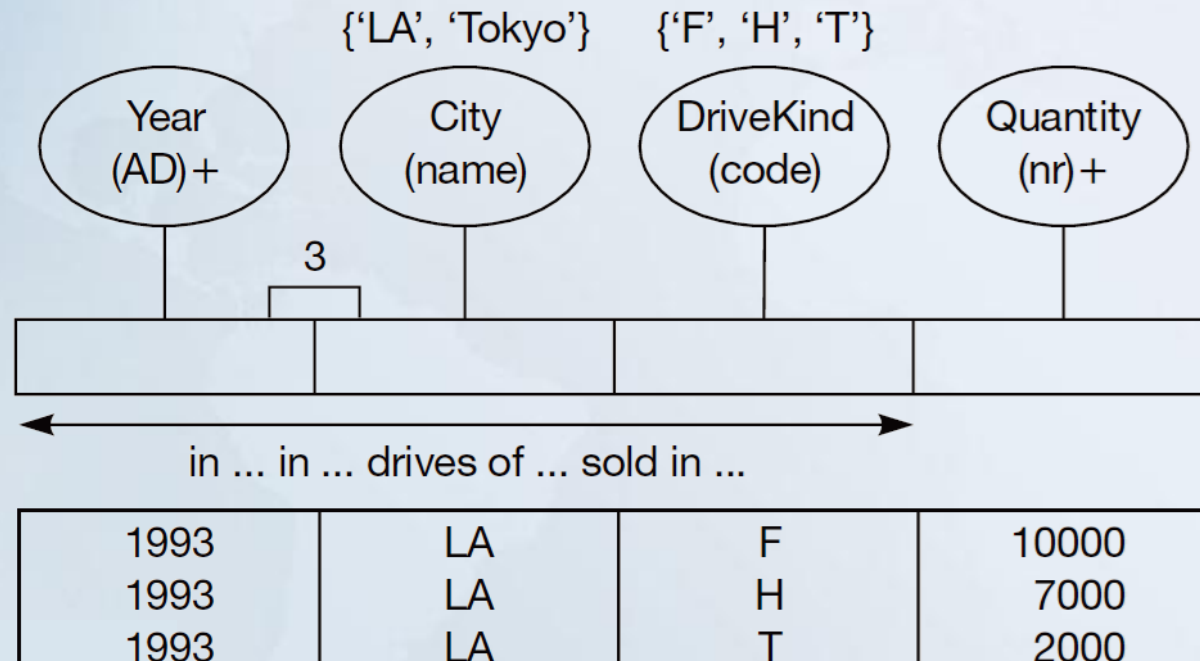
Each member of $\text{pop}(r)$ occurs there exactly n times.

n must be a positive integer.



If $n = 1$, this is equivalent to a uniqueness constraint

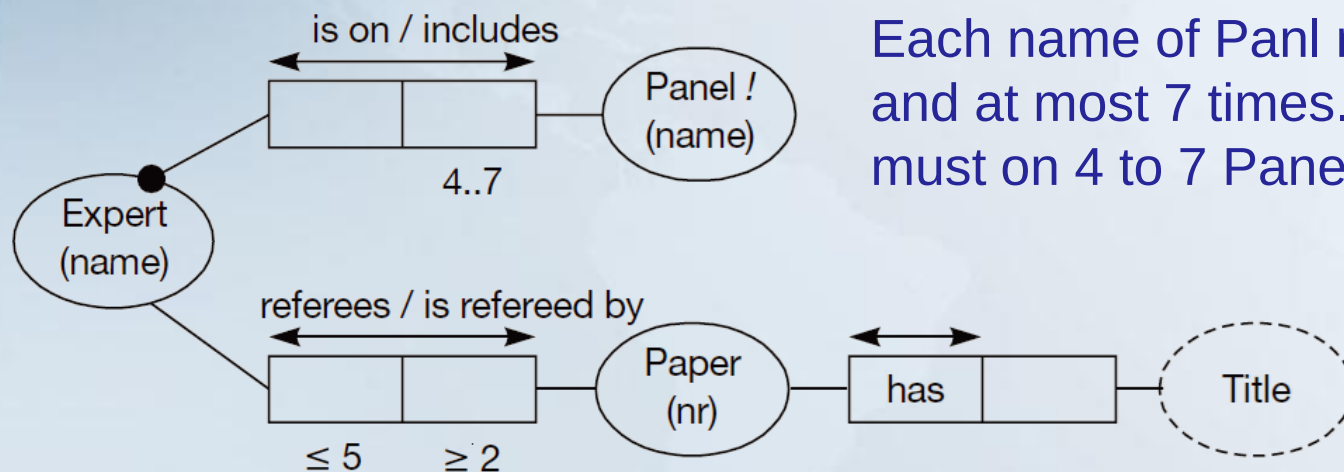
Compound Frequency Constraint



The values of (Year and City) must occur exactly three times

Ranged Frequency Constraint

Examples of minimum and maximum frequency constraints.



Each name of Panel must occur at least 4 and at most 7 times. That is, each Expert must on 4 to 7 Panels

Each Expert can referee 5 papers

Each Paper can be refereed by at least two Experts.

Discussion

Summarize what you learned? And what you think about it?

Compare what you learned with EER and UML?

Questions & Suggestions?

References

1. Terry Halpin, Tony Morgan: **Information Modeling and Relational Databases**, Second Edition. Second Edition. The Morgan Kaufmann Series in Data Management Systems. ISBN: 0123735688